Quantification: 872_wt_hp_z04
==============================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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| measure | value |
|---|---|
| lacuna count | 13 |
| interior lacuna count | 10 |
| frame-edge lacuna count | 3 |

Interior lacunae only: mean, median and sample SD (n = 10)
----------------------------------------------------------

| measure | kind | mean | median | sd | units |
|---|---|---|---|---|---|
| area_px2 | lacuna shape | 2101.33 | 2144 | 927.448 | px^2 |
| major_axis_length_px | lacuna shape | 80.9142 | 76.9066 | 19.4155 | px |
| minor_axis_length_px | lacuna shape | 35.0904 | 34.0354 | 9.2461 | px |
| aspect_ratio | lacuna shape | 2.3403 | 2.5083 | 0.4528 | unitless |
| eccentricity | lacuna shape | 0.8904 | 0.9171 | 0.0524 | unitless |
| solidity | lacuna shape | 0.7707 | 0.7683 | 0.0678 | unitless |
| roots_count | headline | 15.1 | 15.5 | 4.8408 | count |
| ring_length_r30_px | headline | 680.1 | 762 | 194.984 | px |
| ring_length_r60_px | reported | 1606.5 | 1752.5 | 395.107 | px |
| owned_length_px | ownership-dependent | 3376.99 | 3292.33 | 1279.95 | px |
| edge_count | ownership-dependent | 219 | 210.5 | 88.7994 | count |
| mean_edge_length_px | ownership-dependent | 15.6534 | 15.8783 | 1.6297 | px |
| perimeter_px | normalised | 254.63 | 264.764 | 86.6163 | px |
| ring_area_r30_px2 | normalised | 9022 | 9101.5 | 2028.2 | px^2 |
| ring_area_r60_px2 | normalised | 22120.7 | 22141.5 | 4248.88 | px^2 |
| in_frame_fraction_r30 | normalised | 1 | 1 | 0 | unitless |
| in_frame_fraction_r60 | normalised | 1 | 1 | 0 | unitless |
| ring_density_r30 | normalised | 0.0752663 | 0.0790986 | 0.0121753 | px^-1 |
| ring_density_r60 | normalised | 0.0731669 | 0.0744297 | 0.0139928 | px^-1 |
| roots_per_100px_perimeter | normalised | 6.1064 | 6.1795 | 1.2747 | per 100 px |
| ring_attached_length_r30_px | network v2 | 648.7 | 712.5 | 202.532 | px |
| ring_attached_length_r60_px | network v2 | 1492.2 | 1697.5 | 460.387 | px |
| ring_length_w_r30_px | network v2 | 778.366 | 874.047 | 224.674 | px |
| ring_length_w_r60_px | network v2 | 1830.06 | 2000.51 | 464.498 | px |
| sholl_crossings_r10 | network v2 | 14.5 | 15 | 4.6007 | count |
| sholl_crossings_r20 | network v2 | 16.1 | 16.5 | 4.6774 | count |
| sholl_crossings_r30 | network v2 | 20.9 | 21 | 6.6408 | count |
| loop_count | loops, report only | 35.3 | 33.5 | 22.9495 | count |
| loop_edge_share | loops, report only | 0.5552 | 0.6115 | 0.1394 | share |
| ring_width_mean_r30_px | width | 7.2107 | 7.2026 | 0.4206 | px |

Per field
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| measure | value | units |
|---|---|---|
| analysed_area_px2 | 877862 | px^2 |
| total_skeleton_length_px | 36421 | px |
| canalicular_length_density_per_px | 0.0414883 | px^-1 |
| junction_count | 1324 | count |
| junction_density_per_px2 | 0.00150821 | px^-2 |
| lacuna_count | 13 | count |

| | | |
|---|---|---|
| lacunae_per_px2 | 1.481e-05 | px^-2 |
| skeleton_component_count | 115 | count |
| mean_component_length_px | 316.704 | px |
| median_component_length_px | 12 | px |
| field_length_density_w_per_px | 0.0470877 | px^-1 |
| field_density_without_flagged_per_px | 0.0366175 | px^-1 |
| field_density_in_roi_per_px | | px^-1 |
| unattached_length_px | 43 | px |
| unattached_component_count | 3 | count |
| canalicular_length_density_attached_per_px | 0.0414883 | px^-1 |
| field_length_density_with_unattached_per_px | 0.0415373 | px^-1 |
| excluded_area_px2 | 143103 | px^2 |
| analysed_field_area_px2 | 905473 | px^2 |
| loop_count | 487 | count |
| loop_edge_share | 0.6139 | share |
| edge_count | 2748 | count |

Canalicular width, the new measure

| measure | value | units |
|---|---|---|
| width_mean_px | 7.7453 | px |
| width_median_px | 8 | px |
| width_p10_px | 4 | px |
| width_p90_px | 10.7703 | px |
| width_pixels_used | 37238 | count |

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

| lacuna_id | area_px2 | major_axis_length_px | minor_axis_length_px | aspect_ratio | eccentricity | solidity | orientation_rad | centroid_row_px | centroid_col_px | on_border |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 3025 | 100.943 | 44.409 | 2.273 | 0.898 | 0.65 | -1.4641 | 211.119 | 126.158 | False |
| 2 | 2144 | 87.4647 | 34.0354 | 2.5698 | 0.9212 | 0.7528 | -1.2557 | 226.871 | 777.66 | False |
| 3 | 2841 | 103.213 | 37.4578 | 2.7554 | 0.9318 | 0.8351 | -1.3658 | 276.205 | 420.61 | False |
| 4 | 3125 | 68.6713 | 60.8024 | 1.1294 | 0.4648 | 0.909 | 0.622 | 332.681 | 995.111 | True |
| 5 | 3954 | 114.494 | 48.2385 | 2.3735 | 0.9069 | 0.7517 | -1.4049 | 393.248 | 55.5005 | True |
| 6 | 3573 | 94.8523 | 52.0807 | 1.8213 | 0.8358 | 0.8284 | -1.4851 | 395.903 | 280.417 | False |
| 7 | 1409 | 76.4593 | 28.6406 | 2.6696 | 0.9272 | 0.7683 | -1.5183 | 392.216 | 831.581 | False |
| 8 | 2226 | 76.9066 | 39.7827 | 1.9332 | 0.8558 | 0.7987 | -1.558 | 506.567 | 185.868 | False |
| 9 | 1503 | 75.7312 | 26.0285 | 2.9095 | 0.9391 | 0.8628 | -1.3051 | 568.045 | 702.943 | False |
| 10 | 1123 | 61.8357 | 30.8107 | 2.007 | 0.867 | 0.6785 | -1.3397 | 588.985 | 20.1576 | True |
| 11 | 628 | 38.8866 | 23.9678 | 1.6225 | 0.7875 | 0.7235 | -1.4785 | 619.033 | 332.451 | False |
| 12 | 1563 | 73.771 | 29.411 | 2.5083 | 0.9171 | 0.717 | -1.5252 | 651.639 | 225.151 | False |
| 13 | | | | | | | | 264.441 | 629.968 | False |

Network per lacuna

| lacuna_id | roots_count | ring_length_r30_px | ring_length_r60_px | owned_length_px | edge_count | mean_edge_length_px | ring_width_mean_r30_px |
|---|---|---|---|---|---|---|---|
| 1 | 23 | 955 | 1991 | 3217.05 | 203 | 15.8475 | 7.1991 |
| 2 | 19 | 770 | 1744 | 2739.41 | 167 | 16.4036 | 7.1819 |
| 3 | 14 | 754 | 1880 | 3468.16 | 218 | 15.909 | 7.206 |
| 4 | 13 | 358 | 852 | 1304.12 | 86 | 15.1641 | 7.5825 |

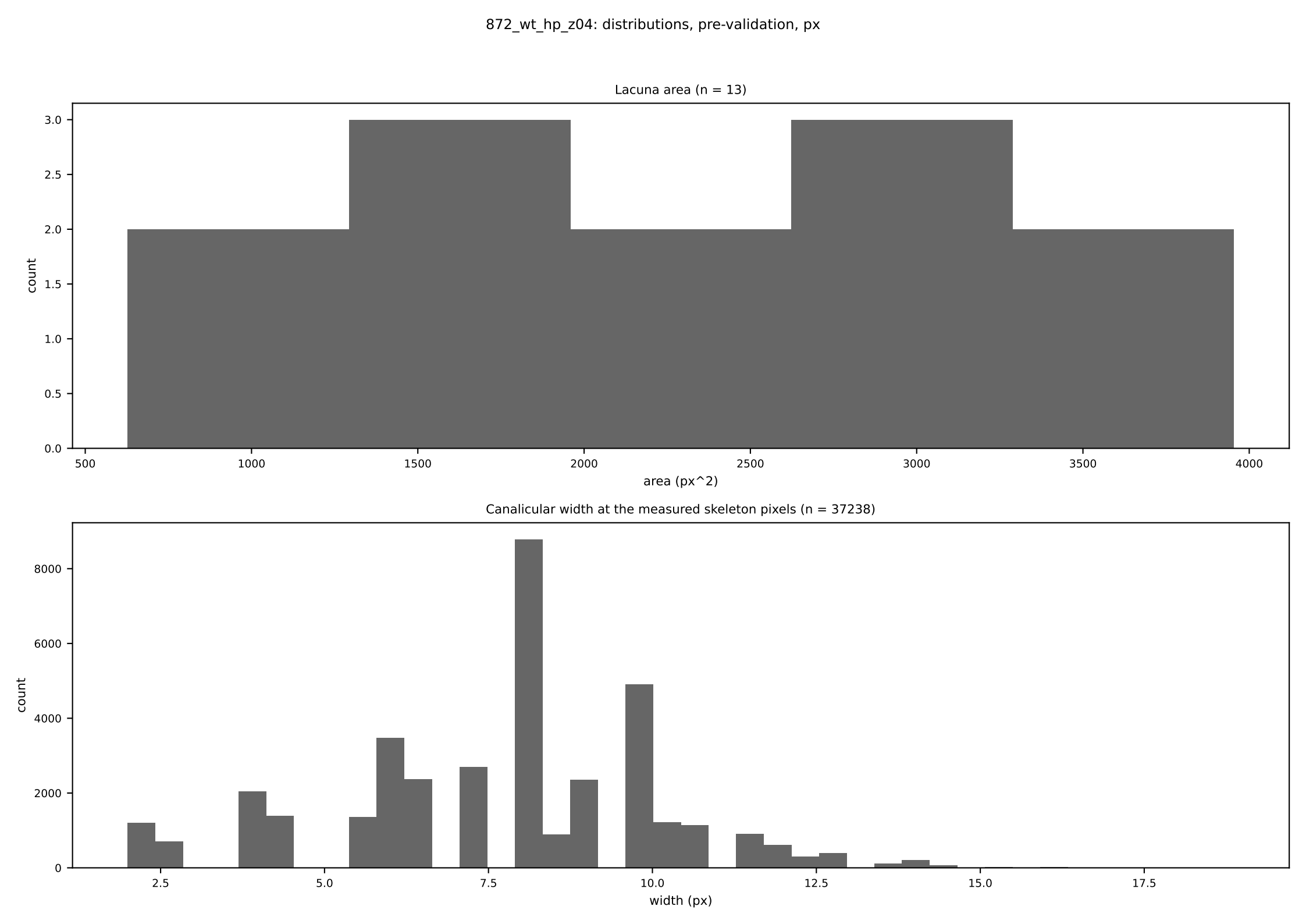
| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 5 | 17 | 607 | 1362 | 1370.58 | 94 | 14.5807 | 7.3359 |
| 6 | 17 | 874 | 2140 | 4649.21 | 306 | 15.1935 | 7.846 |
| 7 | 12 | 409 | 1137 | 2198.56 | 126 | 17.4489 | 6.4577 |
| 8 | 20 | 783 | 1870 | 2108.23 | 171 | 12.3288 | 7.3483 |
| 9 | 9 | 506 | 1063 | 2047.17 | 113 | 18.1165 | 6.8757 |
| 10 | 12 | 482 | 1240 | 2316.1 | 155 | 14.9426 | 6.8827 |
| 11 | 11 | 499 | 1252 | 6156.17 | 413 | 14.906 | 6.8692 |
| 12 | 17 | 802 | 1761 | 3367.6 | 236 | 14.2695 | 7.3103 |
| 13 | 9 | 449 | 1227 | 3818.33 | 237 | 16.1111 | 7.8128 |

Normalised per lacuna

| lacuna_id | perimeter_px | ring_area_r30_px2 | ring_area_r60_px2 | in_frame_fraction_r30 | in_frame_fraction_r60 | ring_density_r30 | ring_density_r60 | roots_per_100px_perimeter |
|---|---|---|---|---|---|---|---|---|
| 1 | 425.345 | 11873 | 28650 | 1 | 1 | 0.0804346 | 0.0694939 | 5.4074 |
| 2 | 273.321 | 9846 | 24205 | 1 | 1 | 0.0782044 | 0.0720512 | 6.9515 |
| 3 | 288.978 | 10581 | 26042 | 1 | 1 | 0.0712598 | 0.0721911 | 4.8447 |
| 4 | 236.551 | 6707 | 15888 | 0.711 | 0.6585 | 0.0533771 | 0.0536254 | 5.4956 |
| 5 | 432.996 | 10444 | 22425 | 0.8508 | 0.7745 | 0.0581195 | 0.0607358 | 3.9261 |
| 6 | 315.191 | 11229 | 25289 | 1 | 1 | 0.0778342 | 0.0846218 | 5.3936 |
| 7 | 256.208 | 8359 | 21924 | 1 | 1 | 0.0489293 | 0.051861 | 4.6837 |
| 8 | 273.764 | 9314 | 21493 | 1 | 1 | 0.084067 | 0.0870051 | 7.3055 |
| 9 | 204.445 | 8382 | 22359 | 1 | 1 | 0.0603675 | 0.0475424 | 4.4022 |
| 10 | 216.752 | 5820 | 13951 | 0.7153 | 0.655 | 0.0828179 | 0.0888825 | 5.5363 |
| 11 | 148.61 | 6134 | 15204 | 1 | 1 | 0.0813499 | 0.0823467 | 7.4019 |
| 12 | 243.386 | 8889 | 20037 | 1 | 1 | 0.0902239 | 0.0878874 | 6.9848 |
| 13 | 117.054 | 5613 | 16004 | 1 | 1 | 0.0799929 | 0.0766683 | 7.6888 |

Network v2 per lacuna

| lacuna_id | ring_attached_length_r30_px | ring_attached_length_r60_px | ring_length_w_r30_px | ring_length_w_r60_px | sholl_crossings_r10 | sholl_crossings_r20 | sholl_crossings_r30 |
|---|---|---|---|---|---|---|---|
| 1 | 930 | 1748 | 1101.89 | 2270.4 | 22 | 18 | 26 |
| 2 | 693 | 1661 | 886.867 | 1995.75 | 17 | 16 | 26 |
| 3 | 732 | 1812 | 863.654 | 2153.24 | 18 | 19 | 22 |
| 4 | 308 | 675 | 399.635 | 969.708 | 7 | 12 | 12 |
| 5 | 564 | 1127 | 706.683 | 1562.71 | 14 | 14 | 14 |
| 6 | 854 | 2137 | 998.767 | 2464.55 | 19 | 25 | 31 |
| 7 | 342 | 940 | 458.605 | 1271.38 | 9 | 12 | 13 |
| 8 | 779 | 1853 | 884.441 | 2140.3 | 14 | 17 | 27 |
| 9 | 459 | 740 | 574.132 | 1183.22 | 11 | 8 | 15 |
| 10 | 475 | 1239 | 544.676 | 1418.12 | 12 | 12 | 14 |
| 11 | 466 | 1147 | 565.132 | 1416.15 | 10 | 14 | 20 |
| 12 | 784 | 1734 | 919.583 | 2005.28 | 16 | 19 | 18 |
| 13 | 448 | 1150 | 530.588 | 1400.36 | 9 | 13 | 11 |


872_wt_hp_z04: distributions, pre-validation, px
Lacuna area (n = 13)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 37238)
count
width (px)